

OFFAIR AI Test Knowledge Base Document

Document ID: SOP-MECH-042 Department: Maintenance Classification: Internal Allowed Roles: Engineer, Technician

Standard Operating Procedure (SOP)

Pump P-104 Maintenance & Inspection

1. Purpose

This Standard Operating Procedure defines the preventive maintenance and inspection process for Centrifugal Pump P-104, used in the refinery cooling water circulation system. The objective is to ensure safe operation, reduce unexpected downtime, and maintain equipment reliability.

2. Equipment Information

Parameter	Value
Asset ID	P-104
Equipment Type	Centrifugal Pump
Department	Mechanical Maintenance
Operating Medium	Cooling Water
Rated Pressure	8 bar
Motor Rating	55 kW

3. Routine Inspection Checklist

Inspect the following parameters before every shift.

Inspection Item	Acceptable Range
Bearing Temperature	Below 80°C
Pump Vibration	Below 4.5 mm/s
Seal Leakage	Less than 5 drops/min
Discharge Pressure	7.8–8.2 bar
Motor Current	92–108 A

If any value exceeds the acceptable range, immediately notify the maintenance engineer.

4. Maintenance Procedure

1. Stop Pump P-104 using the approved shutdown sequence.
2. Isolate the electrical supply.
3. Apply Lock-Out Tag-Out (LOTO) before opening the coupling guard.

4. Inspect bearings for abnormal wear or overheating.
5. Measure shaft alignment using a dial indicator.
6. Replace bearings if temperature exceeds 80°C.
7. Record observations in the maintenance logbook.
8. Restart the pump only after supervisor approval.

5. Critical Safety Instructions

- Always follow the LOTO procedure before maintenance.
- Never remove the coupling guard while the motor is energized.
- Use insulated gloves and safety goggles during inspection.
- If vibration exceeds 4.5 mm/s, stop the pump immediately.
- Report repeated overheating incidents under Asset ID P-104.

6. Common Failure Symptoms

Symptom	Possible Cause
High bearing temperature	Bearing wear or lubrication failure
Abnormal vibration	Shaft misalignment or impeller damage
Low discharge pressure	Blocked suction line
Excessive seal leakage	Mechanical seal failure

7. Engineering Notes

Any inspection report containing terms such as:

- Pump overheating
- High bearing temperature
- Abnormal vibration
- P-104
- Bearing wear

should retrieve this SOP as the primary maintenance reference within the enterprise knowledge base.

8. Approval

Prepared By: Mechanical Engineering Division

Approved By: Maintenance Manager

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